

MINISTRY OF EDUCATION AND RESEARCH



TECHNICAL UNIVERSITY
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Assignment *number*

Software Design

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Group: *group*

FACULTY OF AUTOMATION AND COMPUTER SCIENCE

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1 Problem definition

This section describes the general objective of the project. You can extract the problem definition directly from the assignment document and complete it with anything else you consider necessary to explain what your application will do.

This section should also introduce the following sections (the development phases) and describe what each of them is concerned with.

At the end you should include an additional paragraph listing the deliverables for this assignment.

2 Analysis phase - Use cases & requirements

2.1 Functional requirements

Functional requirements describe the operations that the system must perform.

These requirements define the services the system provides and the expected system behavior. You usually derive them from the problem statement.

Examples of functional requirements include:

- The system must allow users to authenticate.
- The system must allow users to create, update and delete records.
- The system must allow users to search or filter data.

For this laboratory, non-functional requirements are out of scope, but you can optionally exercise specifying some that are related to the assignment topic (such as security requirements, scalability, performance etc.).

2.2 Use cases

Use cases describe in text the interactions between actors and the system in order to accomplish specific tasks.

Each use case should include:

- The actor initiating the interaction
- A description of the task being performed
- The result of a successful interaction

Do not forget to include a use case diagram here!

An example of embedding an image in a LaTeX document:



Figure 1: A cat for you

3 Design phase

3.1 Architecture overview

This subsection presents the high-level architecture of the application. It should describe how the system is divided into major components such as client applications, backend services and the database.

The chosen architectural style such as client-server or layered architecture may be discussed here, with the advantages and disadvantages it brings.

3.2 Entity-relationship diagram

This subsection describes the database schema used by the application.

The entity-relationship diagram should illustrate the structure of the database and the relationships between entities.

Each table should be described briefly, including:

- Table purpose
- Important attributes
- Relationships with other tables

3.3 Component X

This subsection is an example to describe one of the main components of the system. This could represent the server application, a backend service or a specific module of the system.

The description should include the responsibilities of the component and how it interacts with other parts of the system.

3.4 Component Y

This subsection describes another component of the system, such as the client application or an additional service layer.

Explain its role in the architecture and the type of functionality it provides.

3.5 Design patterns

This subsection explains the design patterns used in the system.

For each pattern you should describe:

- The category of the pattern (creational, structural, behavioral)
- The problem it solves
- The classes involved in the implementation

3.6 Class diagrams

Class diagrams describe the static structure of the system by presenting classes, attributes, methods and relationships between classes.

This section should include one or more class diagrams illustrating the internal organization of the system components.

3.7 Sequence diagrams

Sequence diagrams illustrate how objects interact during the execution of a particular use case.

They show the order of method calls and message exchanges between system components.

4 Implementation phase - Application

4.1 Tools and technologies

This subsection presents the technology stack used to implement the application.

You should describe the main tools and technologies used during development, such as:

- Programming language
- Database management system
- Frameworks or third-party libraries
- Communication protocols

You should also explain the reasons for choosing these technologies.

4.2 Application structure

This subsection describes the structure of the source code and how the project is organized. For example, you may describe packages, modules, or layers used in the application.

4.3 User manual

Here you should present the graphical user interface of the application and explain how users interact with the system.

The user manual should describe how a user can access the main functionalities of the application depending on their role.

Don't be shy to include screenshots of the interface whenever necessary to illustrate the different features available to users!

5 Testing and validation

This section describes the methods used to verify that the application works correctly.

Testing should include unit testing specific to the programming language/framework you choose!

The goal is to demonstrate that the implemented system satisfies the requirements defined in the analysis phase.